

David Khachatryan - Week 2 Report

What went well

My new partner Burke and I managed to finish the first part during the class time. We contributed to the effort equally, while having great chemistry. I am really excited to continue collaborating with him. Another thing that made our collaboration seamless was the use of the LiveShare VSCode extension. I like using this for group projects, and Burke was really excited to get started on it. This ensured that we are both coding on the project live. For example, for a safety check function, we wrote if conditions simultaneously, which made it twice faster to get it done.

Another thing that went well was setting up the graph generation pipeline. We wrote a quick python script to read a log file and generate the graph for acceleration, integral, and filter pitch/roll. This allowed us to do a lot of test runs until we got the correct offshoot.

What did not go well, why?

We struggled with displaying the correct offshoot. The main bottleneck was moving the IMU correctly. We tended to move it way faster (30 degrees/sec) than it should have been (5-10 degrees/sec). This made it tricky to get the graph offshoot. On the upside, our code was not the issue, which was great.

What will you change for next class

Going forward, I should pay more attention to the way I control the IMU, and subsequently the drone. I haven't flown a drone before, so this week I am planning to fly a beginner friendly HolyStone drone to get a feel for how to control it. Obviously, the commercially available drones have their own flight controllers, which makes them really easy to fly.